Levelling the Field:

A Statistical Analysis on Internet Connection across Europe

Presented By: Luca Stucchi, Stole Donev, Maxime Engels, Tiziano Gallo

Presentation Overview

EU Digital Divide & Digital Decade Targets

Research Questions & Data

Key Findings & Statistical Analysis

Conclusions & Recommendations

The EU Digital Decade 2030

01

Legally binding target

100% gigabit internet for all EU households by 2030.
In 2020 only 13% of regions met even 100 Mbps.

02

Progress is real — but uneven

62% of regions on track by 2022, up from 13%.
But 38% still lag, and speed inequality is growing.

03

National averages hide the real problem

Serbia, France and Czechia have severe within-country speed disparities even in relatively wealthy states.

★ ★ ★ ★ ★ ★
★ ★ ★ ★ ★ ★

100%

gigabit by 2030

EU Digital Decade Target

2022: 62% on track
38% still lagging

European Commission

Policy mandate

Accountable to the European Parliament for the 2030 target

Funding allocation

Directs EU Cohesion Funds — billions for closing the digital divide

Where to invest?

Our analysis tells them which regions to prioritize

Research Questions

RQ1

Are EU regions converging towards the 100 Mbps target?

What structural factors drive the remaining gap?

RQ2

Is internet access really equal across EU regions?

Or does download speed hide a deeper divide?

RQ3

Which regions are most suitable for satellite internet?

Where should the EU prioritize last-mile investment?

Data

Source: Eurostat

Broadband Infrastructure Data

Datasets:

Speed

Household Access

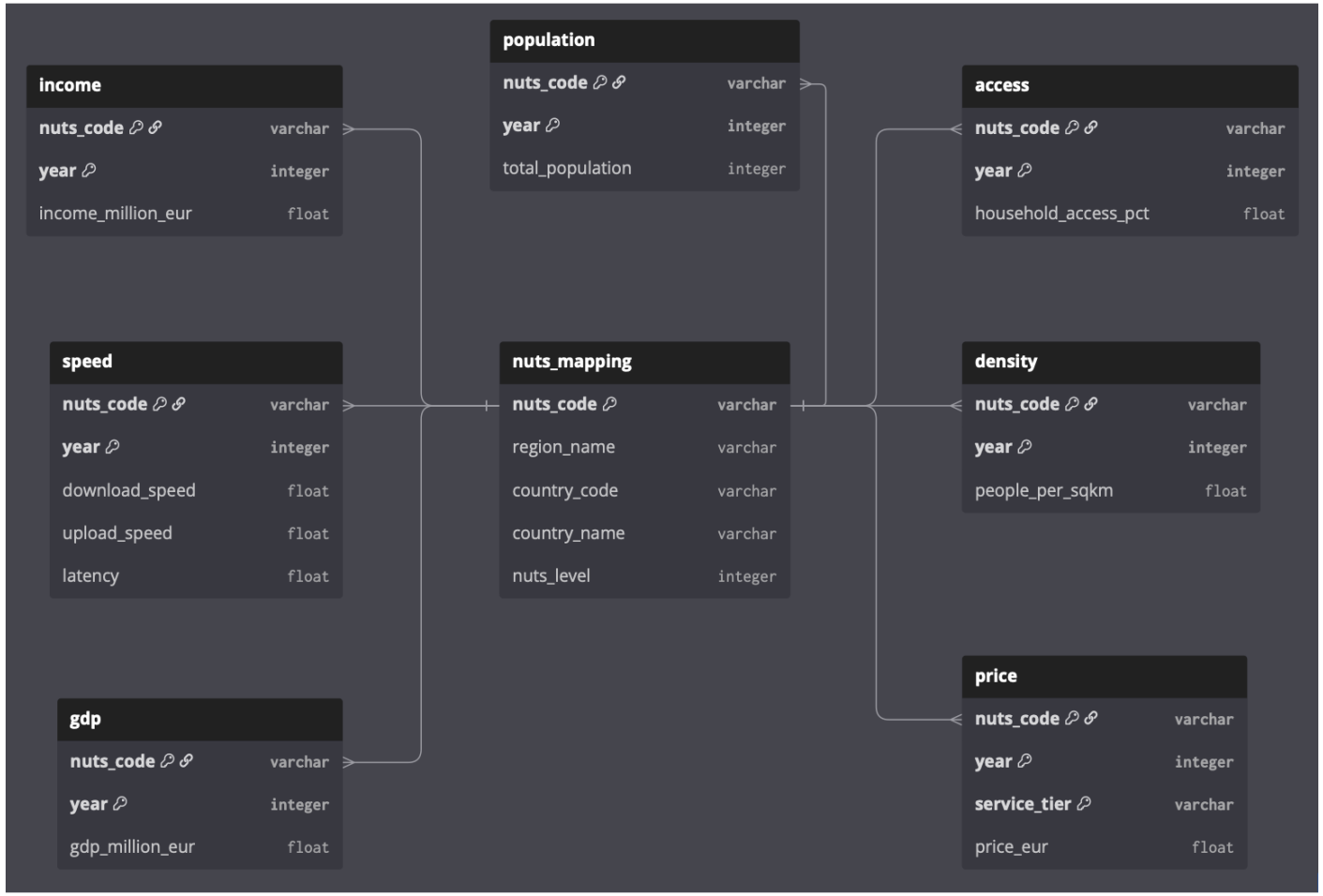
Price

GDP

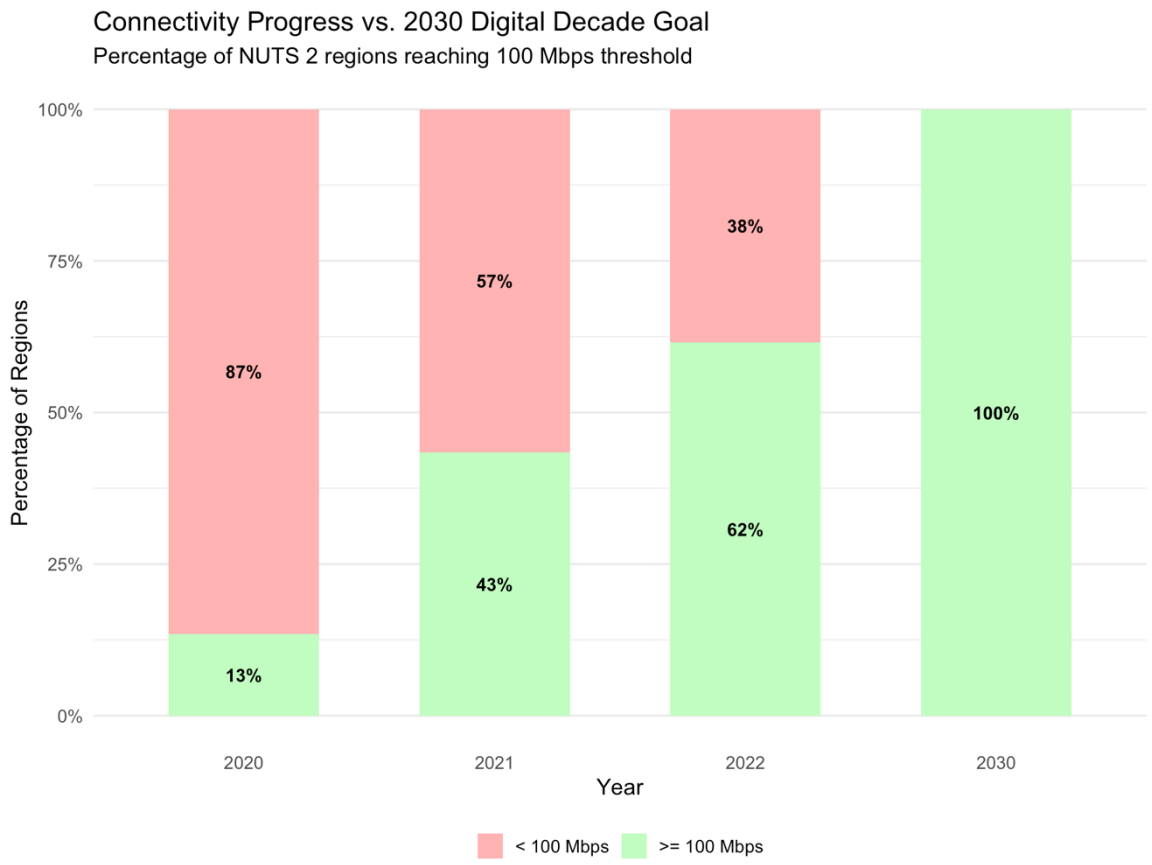
Density

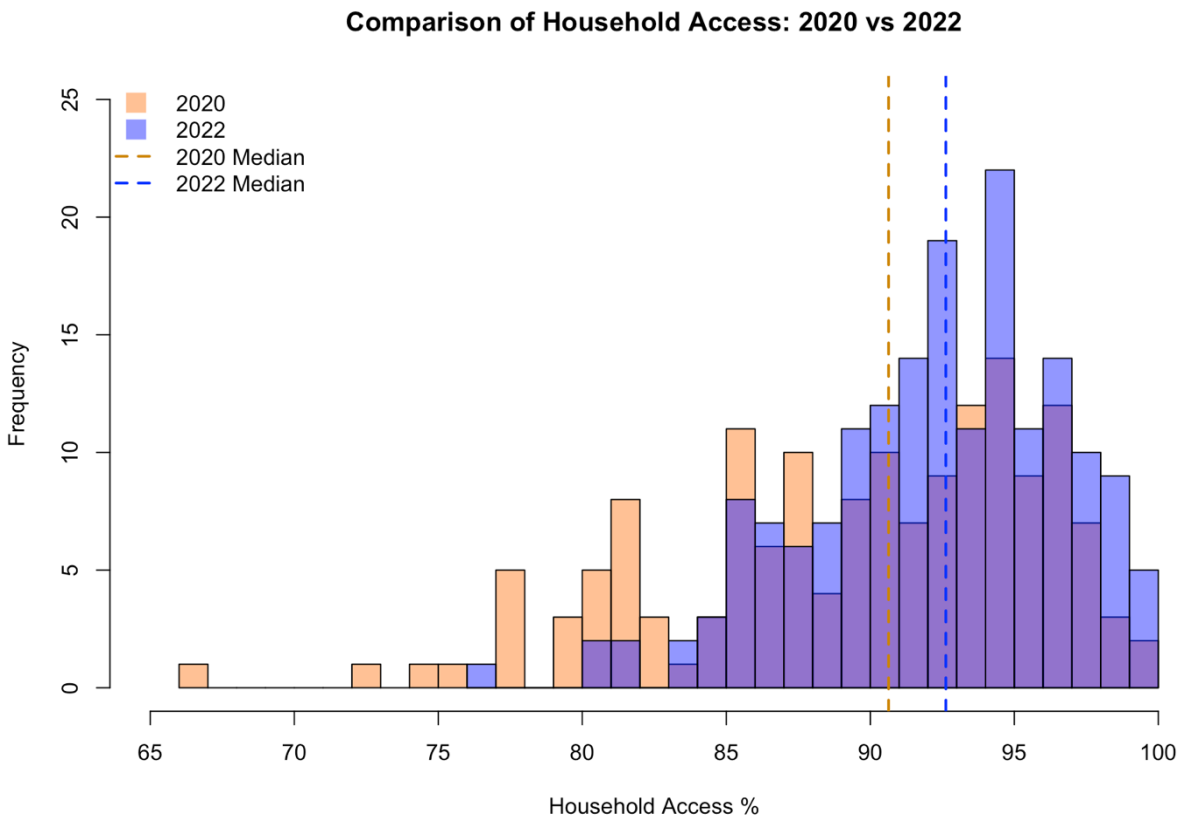
Population

Income



Progress Trend from 2020 – 2022



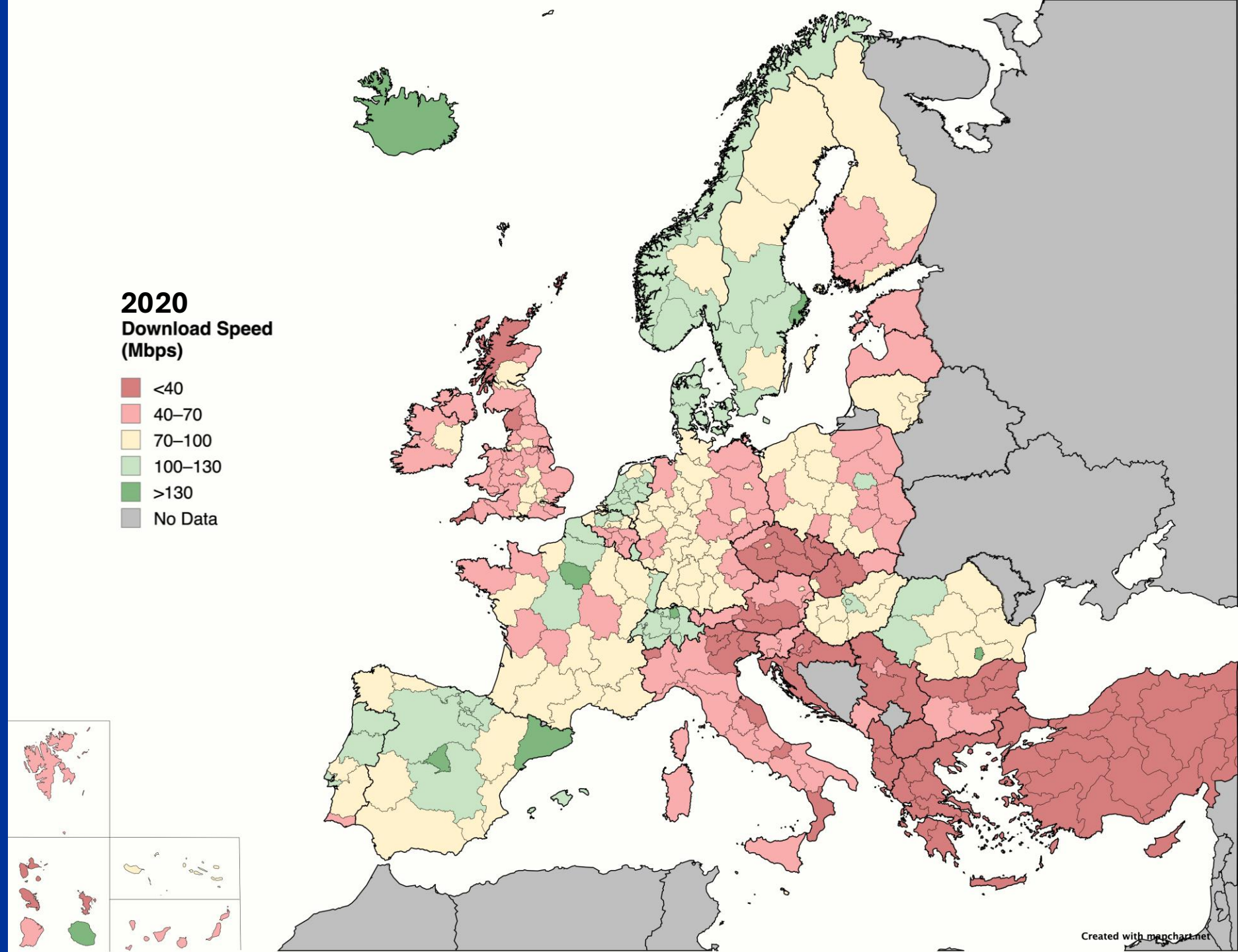


364
Regions

65
Median Speed

315
Regions <100 Mbps

49
Regions ≥100 Mbps

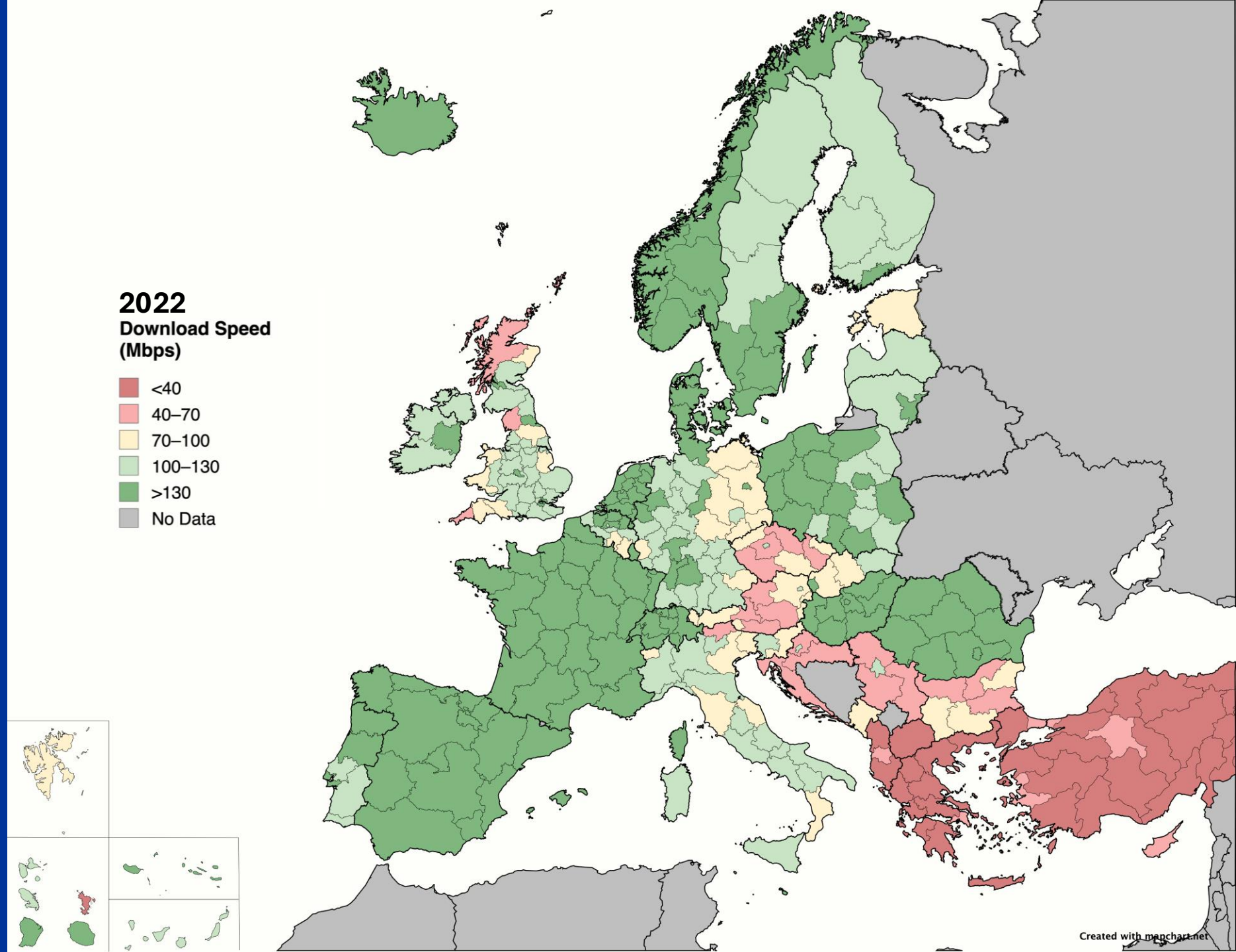


364
Regions

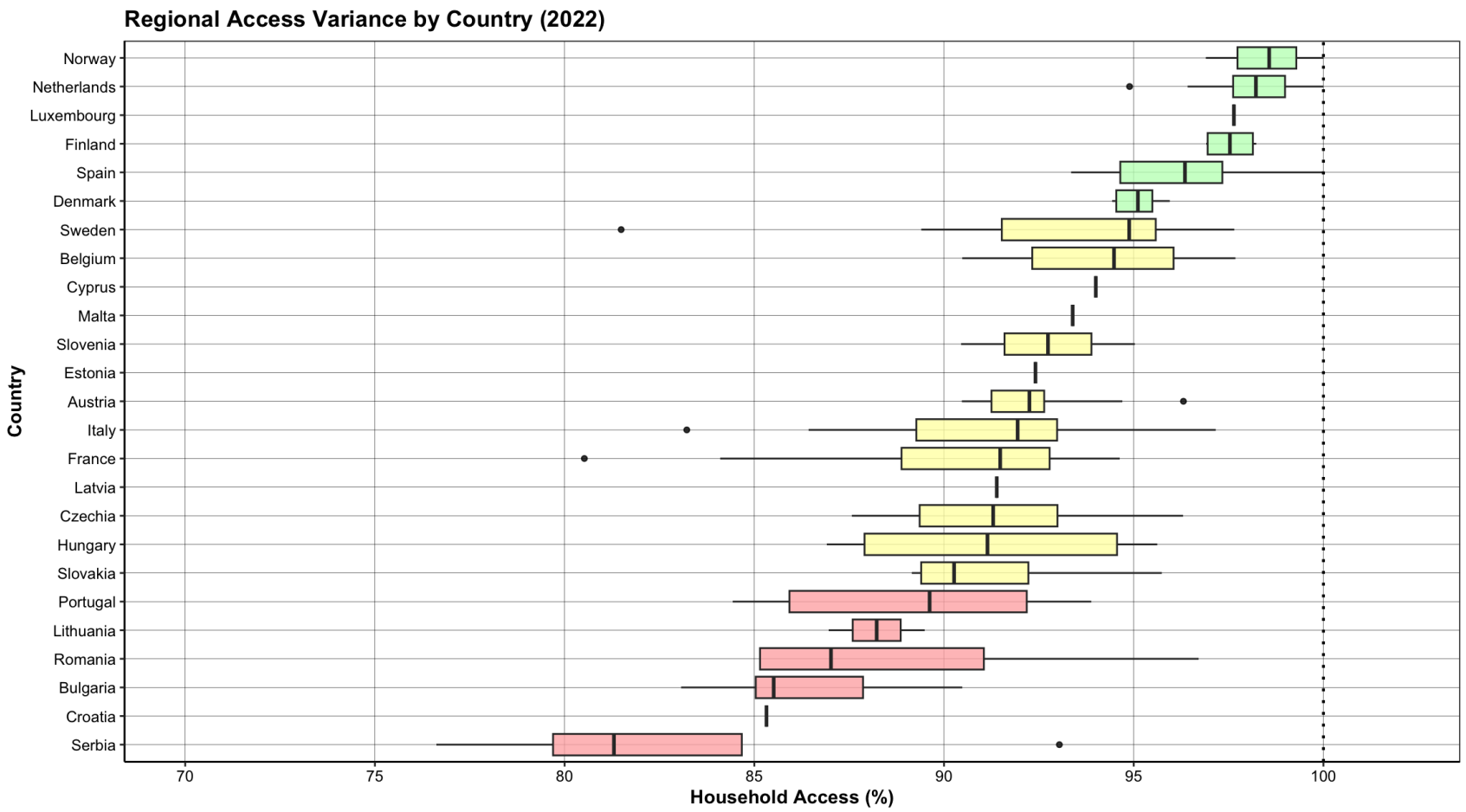
113
Median Speed

140
Regions <100 Mbps

224
Regions ≥100 Mbps



Finding Countries with Low Household Access

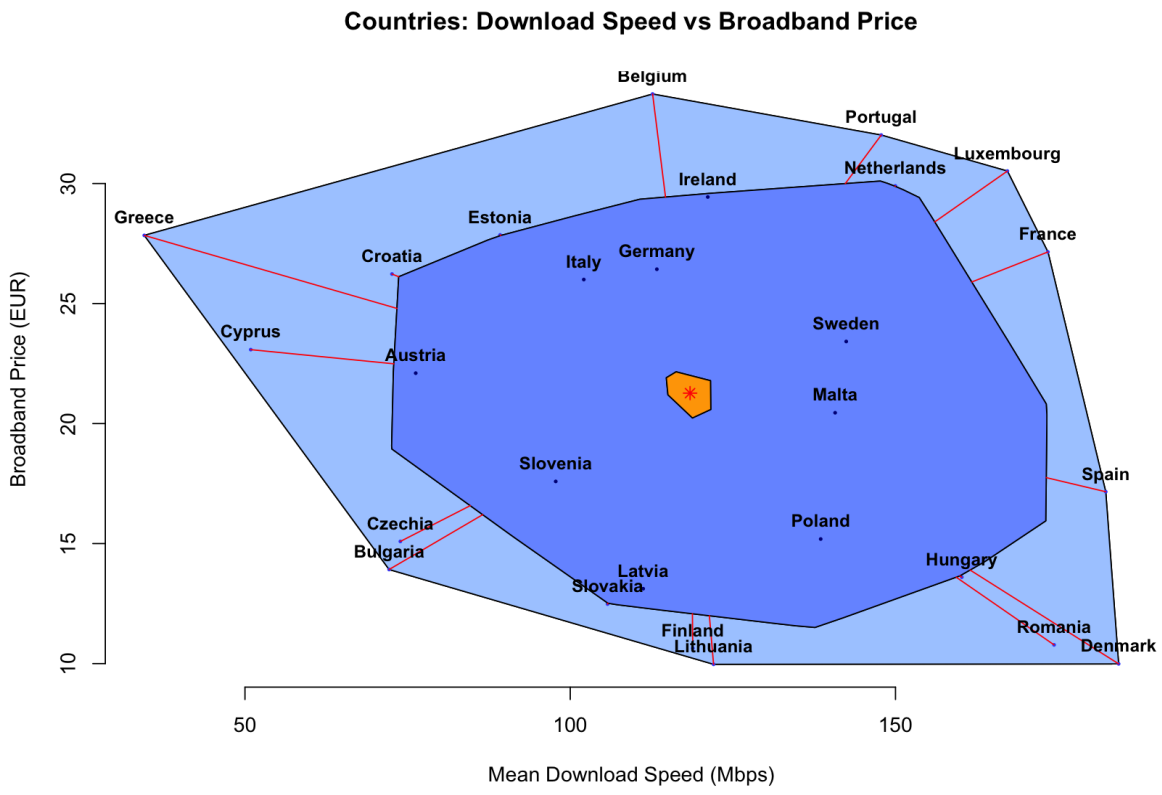


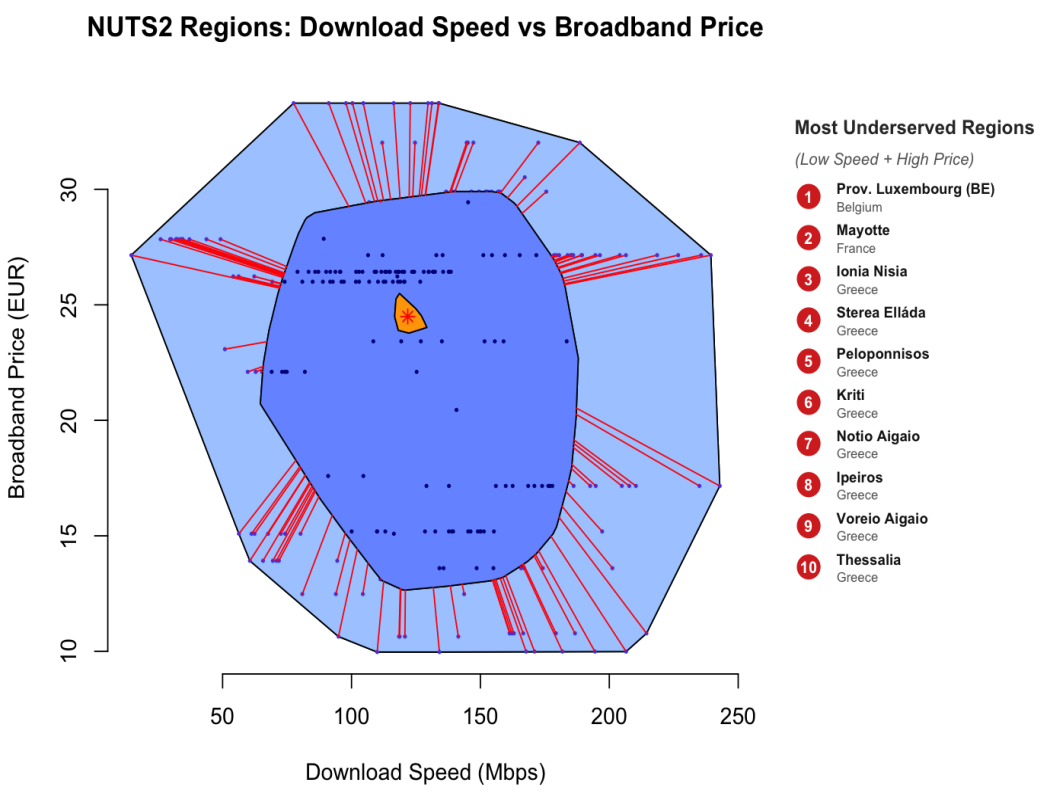
Red Zones

(<90% median internet access)

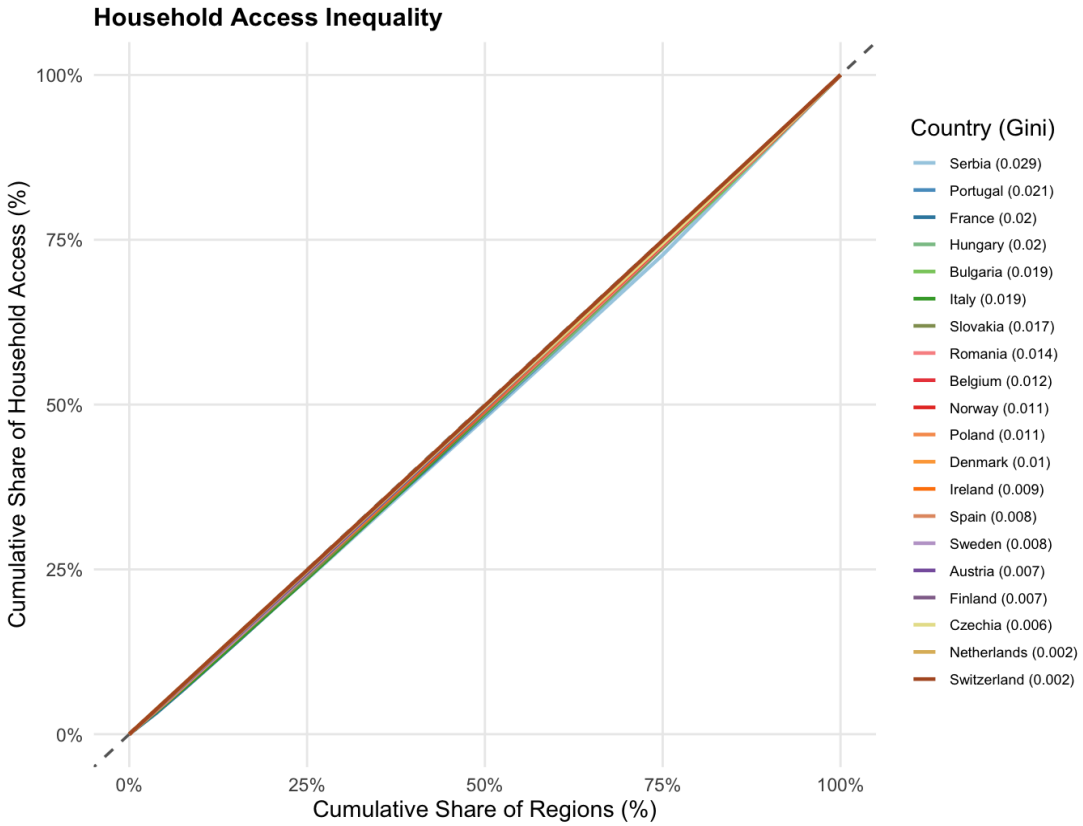
- Serbia
- Croatia
- Bulgaria
- Romania
- Lithuania
- Portugal

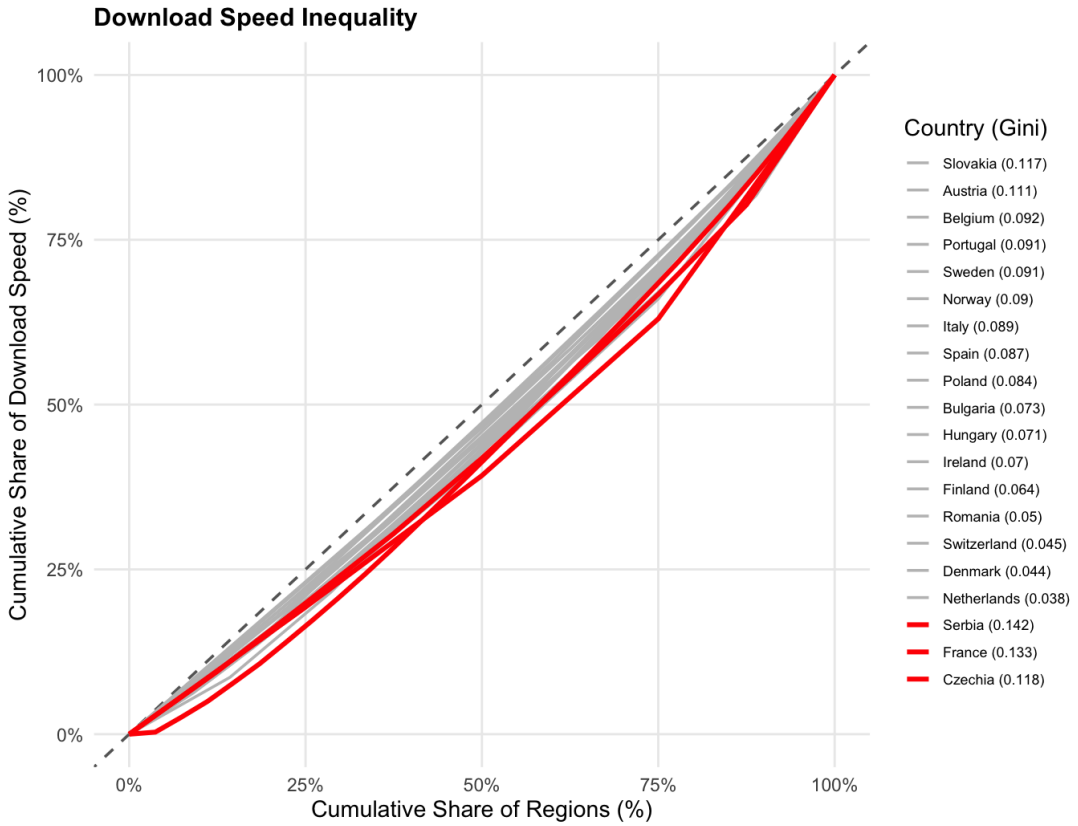
Countries and Regions in Need of Help





Gini & Lorenz: Inequalities within a Country

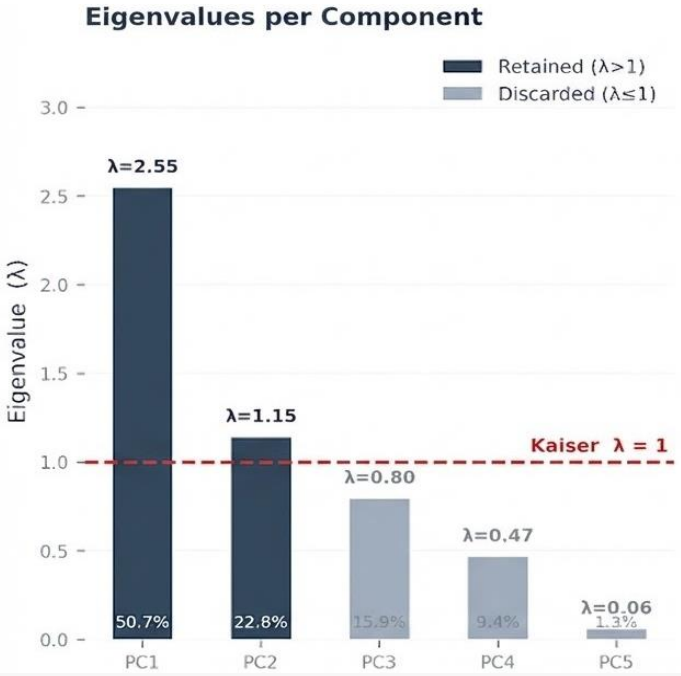


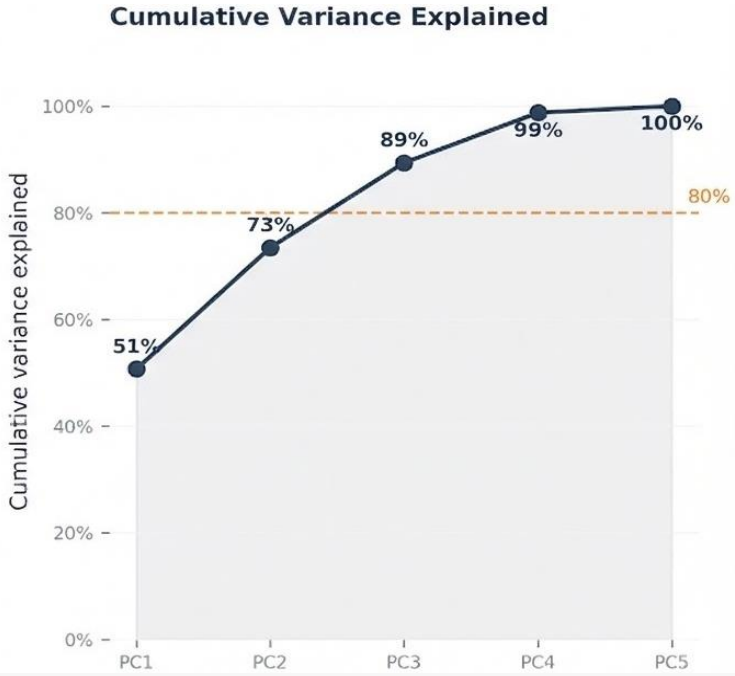


Perfect Equality

Serbia, France, Czechia

PCA



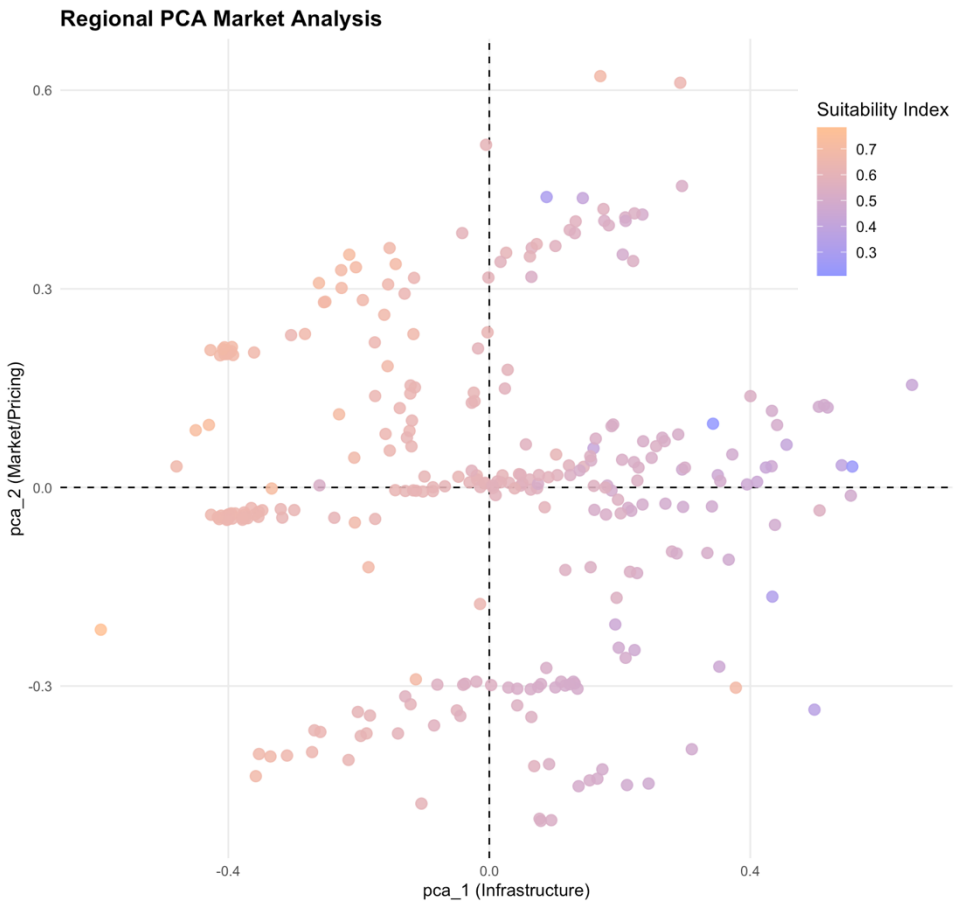


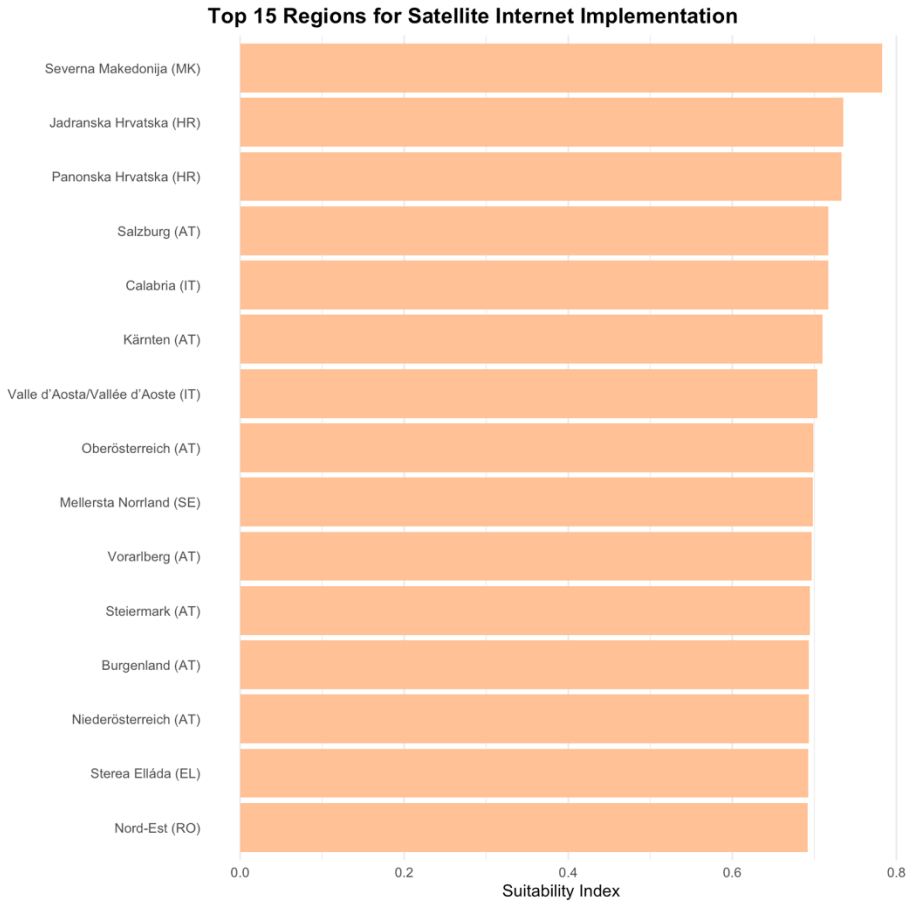
PC1 50.7%
λ=2.55 'rich & connected' composite. All 5 variables load positively

PC2 22.8%
λ=1.15 — Speed & density vs income & GDP. Separates dense/fast/poor from sparse/wealthy

Kaiser Criterion
Retain λ>1 only. PC3–5 discarded — less informative than a single variable

PCA





Regions with low PC1 (infra gap) but positive PC2 (price tolerance) are satellite-optimal. N. Macedonia, Croatia, Calabria and N. Hungary lead the target list

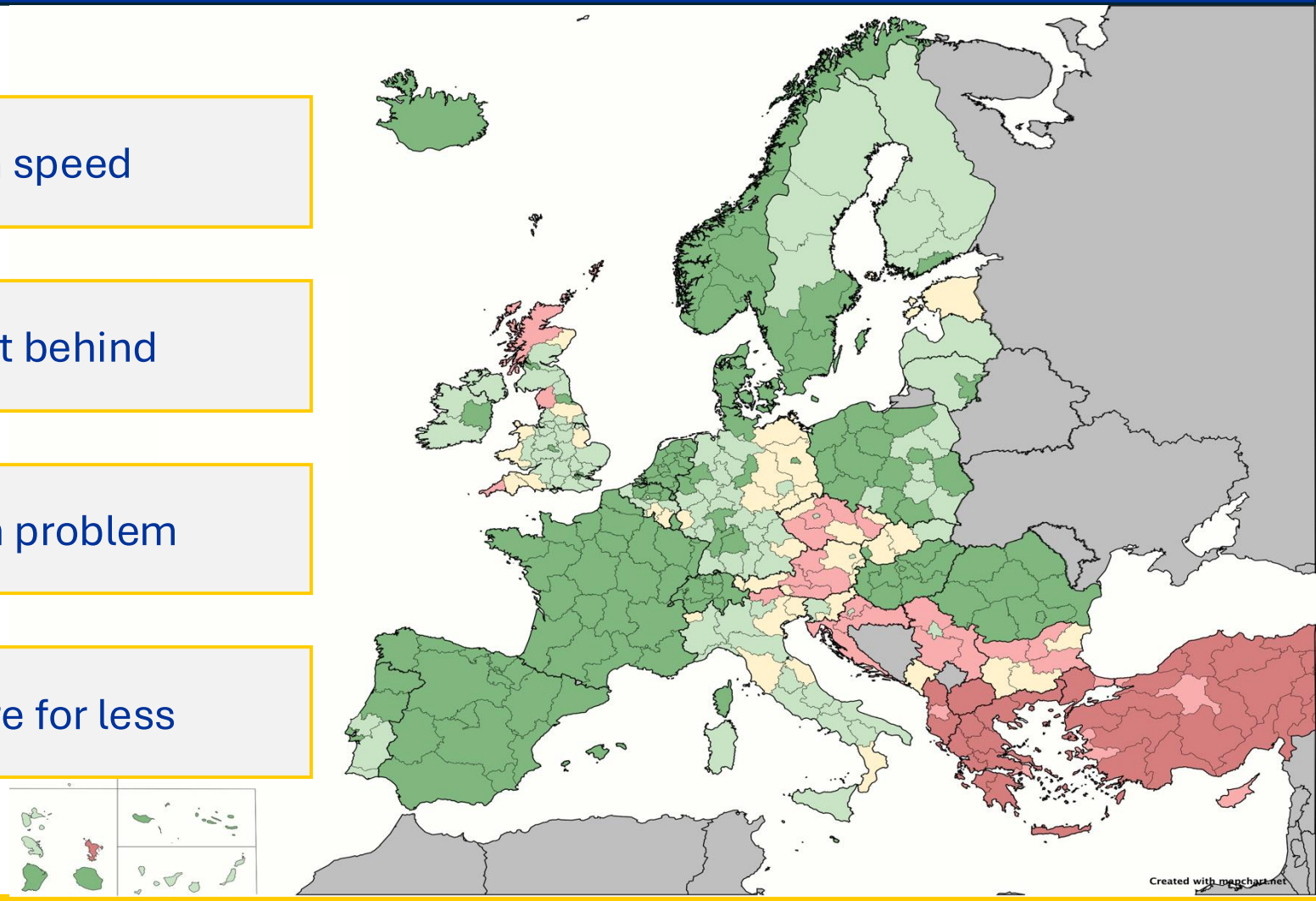
Conclusions

Equal access but not enough speed

South-East Europe is being left behind

Internal inequality is the hidden problem

Some countries are paying more for less



Recommendations

Improve Infrastructure + Reduce Price Gap

1. Target Red-Zone Countries

2. Address Speed Inequality, Not Just Access

3. Pilot Satellite Internet in Top 15 Regions

4. Expand Data Collection to NUTS 3 Level

Limitations

- **Cross-sectional snapshot**
  2022 data only, no long-term causal trend analysis
- **Missing data**
  Price data missing for 10 countries (Serbia, Norway, Switzerland, UK etc.)
- **No satellite penetration data**
  All measurements are terrestrial
- **NUTS 2 aggregation masks sub-regional gaps**
  Unable to pinpoint exact locations/regions

Thank you